ADVANCE SQL QUERY IMPLEMENTATION

AJIO-E COMMERCE DATABASE MANAGEMENT SYSTEM

1. Subqueries and nested queries

1.1 Find products whose price is greater than the average product price.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price > (
  4      SELECT AVG(Price)
  5      FROM Product
  6  );

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
       101
Casual Shirt
       999

       103
Running Shoes
      1499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
```

1.2 Find customer IDs whose order amount is greater than the average order amount.

```
SQL> SELECT Customer_ID
  2  FROM Orders
  3  WHERE Total_Amount > (
  4      SELECT AVG(Total_Amount)
  5      FROM Orders
  6      WHERE Total_Amount > (
  7          SELECT MIN(Total_Amount)
  8          FROM Orders
  9      )
 10  );

CUSTOMER_ID
-----------
           3
           5
```

2. Find products above average price

```
SQL> SELECT Product_Name, Brand, Price
  2  FROM Product
  3  WHERE Price > (
  4      SELECT AVG(Price)
  5      FROM Product
  6  )
  7  ORDER BY Price DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                    PRICE
-------------------------------------------------- ----------
Running Shoes
Puma                                                      1499

Casual Shirt
Roadster                                                   999
```

3. Customers with maximum purchases

```
SQL>
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         COUNT(O.Order_ID) AS Total_Orders
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  9  HAVING COUNT(O.Order_ID) = (
 10      SELECT MAX(Order_Count)
 11      FROM (
 12          SELECT COUNT(Order_ID) AS Order_Count
 13          FROM Orders
 14          GROUP BY Customer_ID
 15      )
 16  );

CUSTOMER_ID FIRST_NAME                     LAST_NAME
----------- ------------------------------ ------------------------------
TOTAL_ORDERS
------------
          1 Pooja                          Kumar
            1

          2 Rahul                          Ravi
            1

          3 Ananya                         Devi
            1


CUSTOMER_ID FIRST_NAME                     LAST_NAME
----------- ------------------------------ ------------------------------
TOTAL_ORDERS
------------
          4 Arjun                          Raj
            1

          5 Priya                          S
            1
```

4. Complex business queries

4.1 Query 1: Customer Purchase Details

```
SQL>
SQL> SELECT C.Customer_ID,
  2         O.Order_ID,
  3         O.Order_Date,
  4         O.Total_Amount,
  5         O.Order_Status,
  6         O.Shipping_Address,
  7         O.Delivery_Date
  8  FROM Customer C
  9  JOIN Orders O
 10  ON C.Customer_ID = O.Customer_ID
 11  ORDER BY O.Total_Amount DESC;

CUSTOMER_ID   ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ---------- --------- ------------ ------------------------------
SHIPPING_ADDRESS
--------------------------------------------------------------------------------
DELIVERY_
---------
         3        103 12-SEP-26         3299 Processing
Coimbatore, Tamil Nadu
17-SEP-26

         5        105 14-SEP-26         2899 Shipped
Hyderabad, Telangana
18-SEP-26

CUSTOMER_ID   ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ---------- --------- ------------ ------------------------------
SHIPPING_ADDRESS
--------------------------------------------------------------------------------
DELIVERY_
---------

         1        101 10-SEP-26         2499 Delivered
Chennai, Tamil Nadu
14-SEP-26

         2        102 11-SEP-26         1799 Shipped
Bangalore, Karnataka
```

4.2 Query 2: Find the highest order amount for each customer

```
SQL>
SQL> SELECT Customer_ID,
  2         MAX(Total_Amount) AS Highest_Order
  3  FROM Orders
  4  GROUP BY Customer_ID
  5  ORDER BY Highest_Order DESC;

CUSTOMER_ID HIGHEST_ORDER
----------- -------------
          3          3299
          5          2899
          1          2499
          2          1799
          4          1499
```

4.3 Query 3: Find customers whose order amount is greater than ₹2000

```
SQL>
SQL> SELECT Customer_ID, Order_ID, Total_Amount
  2  FROM Orders
  3  WHERE Total_Amount > 2000
  4  ORDER BY Total_Amount DESC;

CUSTOMER_ID   ORDER_ID TOTAL_AMOUNT
----------- ---------- ------------
          3        103         3299
          5        105         2899
          1        101         2499
```

4.4 Query 4: Display orders based on their status

```
SQL>
SQL> SELECT Order_Status,
  2         COUNT(*) AS Total_Orders,
  3         SUM(Total_Amount) AS Total_Sales
  4  FROM Orders
  5  GROUP BY Order_Status;

ORDER_STATUS                   TOTAL_ORDERS TOTAL_SALES
------------------------------ ------------ -----------
Delivered                                 2        3998
Shipped                                   2        4698
Processing                                1        3299
```

4.5 Query 5: Find customers who have placed delivered orders

```
SQL>
SQL> SELECT Customer_ID, Order_ID, Total_Amount
  2  FROM Orders
  3  WHERE Order_Status = 'Delivered';

CUSTOMER_ID   ORDER_ID TOTAL_AMOUNT
----------- ---------- ------------
          1        101         2499
          4        104         1499
```

5.Advanced SQL Report

5.1 Overall Sales Report

```
SQL> SELECT
  2      COUNT(ORDER_ID) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE,
  4      ROUND(AVG(TOTAL_AMOUNT),2) AS AVG_ORDER_VALUE,
  5      MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER,
  6      MIN(TOTAL_AMOUNT) AS LOWEST_ORDER
  7  FROM ORDERS;

TOTAL_ORDERS TOTAL_REVENUE AVG_ORDER_VALUE HIGHEST_ORDER LOWEST_ORDER
------------ ------------- --------------- ------------- ------------
           5         11995            2399          3299         1499
```

6. Order Status Report

```
SQL>
SQL> SELECT
  2      Order_Status,
  3      COUNT(*) AS Total_Orders,
  4      SUM(Total_Amount) AS Total_Amount
  5  FROM Orders
  6  GROUP BY Order_Status
  7  ORDER BY Total_Amount DESC;

ORDER_STATUS                   TOTAL_ORDERS TOTAL_AMOUNT
------------------------------ ------------ ------------
Shipped                                   2         4698
Delivered                                 2         3998
Processing                                1         3299
```

7. Customer Purchase Report

```
SQL>
SQL> SELECT
  2      Customer_ID,
  3      COUNT(Order_ID) AS Total_Orders,
  4      SUM(Total_Amount) AS Total_Purchase,
  5      ROUND(AVG(Total_Amount), 2) AS Average_Purchase
  6  FROM Orders
  7  GROUP BY Customer_ID
  8  ORDER BY Total_Purchase DESC;

CUSTOMER_ID TOTAL_ORDERS TOTAL_PURCHASE AVERAGE_PURCHASE
----------- ------------ -------------- ----------------
          3            1           3299             3299
          5            1           2899             2899
          1            1           2499             2499
          2            1           1799             1799
          4            1           1499             1499
```

8. Customer and Order Details Report

```
SQL>
SQL> SELECT
  2      C.Customer_ID,
  3      C.City,
  4      COUNT(O.Order_ID) AS Total_Orders,
  5      SUM(O.Total_Amount) AS Total_Spending
  6  FROM Customer C
  7  JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  GROUP BY C.Customer_ID, C.City
 10  ORDER BY Total_Spending DESC;

CUSTOMER_ID CITY                                 TOTAL_ORDERS TOTAL_SPENDING
----------- ---------------------------------------- ------------ --------------
          3 Coimbatore                                        1           3299
          5 Hyderabad                                         1           2899
          1 Chennai                                           1           2499
          2 Bangalore                                         1           1799
          4 Madurai                                           1           1499
```